TAMIZHARUVI MANIYAN ANNAMALAI

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PROFESSIONAL SUMMARY

A versatile and technically driven Senior Naval Architect with over 7 years of progressive experience in ship design, structural engineering, vessel modification, hydrodynamic consultancy, and maritime digital technology. Currently based in London, UK, with Cybermarine Ltd, leading and delivering high-quality naval architecture and engineering projects for international clients across the offshore, subsea, passenger, and commercial maritime sectors. Proven expertise in end-to-end project execution — from concept design and 3D modelling through to production engineering, classification society approvals, and client delivery. Adept at managing multi-disciplinary engineering scopes, providing technical leadership, and driving design automation aligned with Industry 4.0.

Keywords: Naval Architecture | Vessel Modification Engineering | Structural Design | Hydrodynamic Analysis | Stability | CADMATIC | Rhino 3D | 3D Modelling | Production Engineering | Class Approvals | Decarbonisation

PROFESSIONAL EXPERIENCE

Senior Naval Architect — Marine Technology Department *Oct 2024 – Present*
Cybermarine Ltd, London, United Kingdom

Senior Naval Architect & Team Leader — Detailed Design & Hull Structural Design *Mar 2023 – Aug 2024*
Cybermarine Knowledge Systems Pvt Ltd, Mumbai, India

Naval Architect & Team Leader — Detailed Design & Hull Structural Design *Jan 2021 – Mar 2023*
Cybermarine Knowledge Systems Pvt Ltd, Mumbai, India

Junior Naval Architect *May 2018 – Jan 2021*
Cybermarine Knowledge Systems Pvt Ltd, Mumbai, India

KEY PROJECTS UNDERTAKEN

CSV Boka Northern Ocean — Multi-Scope Vessel Modification Engineering (Boskalis Subsea Services, UK)

Ongoing (Sep 2025 – Present) | Class: Bureau Veritas (BV)

Providing comprehensive modification engineering services to Boskalis Subsea Services for a series of deck reconfiguration and system modification scopes onboard the SPS vessel CSV Boka Northern Ocean, necessitated by the installation of new ROV systems. All modifications are designed to honour the vessel's present statutory and classification compliance envelope. Scopes include:

- Gangway Access Rearrangement (P&S) — Relocation of gangway platform, provision of new deck access openings, taut wire grillage deconstruction, and walkway / handrail design.
- PDU Container Installation — Engineering of 2 x PDU container installations on the starboard side including seafastening arrangement, walkway details, cable tray arrangement, and deck strength verification.

- Main Crane Bucket Relocation & Bumper Protection — Relocation of main crane (400T) hoist line buckets; design of bumper protection systems around midship doghouse, crane access ladder, and refuelling station pedestal.
- Closing Bulwarks & Freeing Port Modification — Design of high closing bulwarks with hinged freeing ports iwo pilot access openings; freeing port area evaluation per ICLL; loadline plan updates.
- Project Office to ROV Control Room Conversion — Division of existing project office into a 5-person project office and an 8-person ROV control room; full fire safety, access/escape, HVAC, and electrical SLD engineering per SPS Code and SOLAS.
- Aft Mooring Fitting Relocation & Addition — Repositioning and addition of mooring bits, bollards, and Panama chocks at the aft port station; structural foundation verification; mooring arrangement, tank capacity plan, and air vent drawing updates.
- Fire Pump Relocation — Engineering of fire pump relocation from aft thruster room to forward store / A&R winch room; seachest sizing, NPSH and dynamics calculations; piping, structural, and electrical workshop drawings.
- Pilot Access Walkway & Whipline Bucket Platform — Structural engineering of a new walkway platform connecting pilot access to accommodation deck, and a dedicated whipline bucket platform with seafastening details.
- Gastight False Deck (Cofferdam) — Battery Room — Design and engineering of a gastight cofferdam above an Ex-rated battery room to enable working-deck welding; structural scantlings, fire and gas detection / suppression modifications, ventilation schematics, and hazardous area plans.
- Helideck Plan View Update — Updated Helideck Approach/OFS Plan View per CAP 437 to reflect revised working-deck layout for the Helideck Information Plate.

All scopes delivered with clash-free FEED 3D local models in IGES format, Class/Flag documentation, and full technical liaison with Bureau Veritas and third parties towards approval.

Hydrodynamic Consultancy — Roll Reduction & Motion Control Studies

DSV Jana 201 — Roll Reduction & Motion Control Consultancy *Feb 2026*

JANA Marine Services Co. LLC, Khobar, Saudi Arabia | Feb 2026

- Independent hydrodynamic analysis of a Diving Support Vessel under construction, exhibiting uncharacteristically high roll and pitch motions in moderate environmental conditions.
- Performed linear diffraction analysis to evaluate Roll RAOs, natural roll/pitch periods, roll stiffness, damping (bilge keel, viscous, nonlinear GZ), and vessel operability including moonpool and heading effects.
- Reviewed hull form, stability characteristics, shipyard hydrodynamic predictions, and loading conditions to establish an independent baseline.
- Developed qualitative recommendations for motion control: alternative loading arrangements affecting GM and roll stiffness, favourable heading selection, bilge keel optimisation, and evaluation of passive stabilisation options (flume tanks, U-tanks, free surface and controlled-flooding tanks).
- Deliverable: Consolidated technical report covering vessel hydrodynamic review and motion stabilisation recommendations.

National Geographic Gemini — Roll Reduction & Motion Comfort Consultancy *Jul 2025*

Lindblad Expeditions, Ecuador | Jul 2025

- Technical consultancy for a passenger expedition vessel experiencing roll resonance — natural roll period tuned to prevailing wave energy, resulting in disproportionate motion amplitudes in calm conditions.
- Developed hydrodynamic (panel, mass, and damping) model; performed linear diffraction analysis to determine Roll RAO, roll stiffness, added mass/inertia, damping coefficients, and free surface effects.
- Formulated alternative loading conditions to de-tune roll resonance: GM variation, roll radius of gyration adjustment, increased drafts, and partial tank filling strategies.
- Evaluated feasibility of utilising existing double bottom full-breadth tanks as passive stabilisers, including tank tuning, free surface effects, and damping moment vs filling level.
- Assessed bilge keel modification and passive/active roll stabilisation options (flume tanks, U-tanks, stabiliser fins).
- Deliverable: Consolidated operating procedure for roll reduction and feasibility study report on roll stabilisation, including bilge keel modification plan.

HYDRA Floating Aquafarm — Production Engineering & Project Management

North Sea | 2021 – 2023

- End-to-end production engineering and project management for HYDRA, a 75m diameter floating marine aquafarm — a first-of-its-kind structure stationed in the North Sea.
- Translated concept design into a full workshop drawing set, assembly drawings, and bill of materials.
- Managed the structural 3D design in CADMATIC, handling the challenges of radial framing geometry — an unconventional configuration rarely encountered in standard shipbuilding.
- Coordinated production deliverables directly with the fabrication team and ensured manufacturing readiness.

Maersk T-Class Vessels — E-Sail (WASP) Decarbonisation — Project Management & Stability Engineering

Ongoing (2025 – Present)

- Led project management and stability-focused technical delivery for Wind-Assisted Ship Propulsion (E-Sail) installation on Maersk T-Class vessels.
- Managed intact stability, damage stability, WASP criteria evaluation, and fatigue assessment for the installation.
- Coordinated inclining experiments, draft surveys, and installation planning for the E-Sail units.
- Responded to classification society comments with technically detailed replies, ensuring successful comment closure and project completion.

Draft Increase Feasibility Studies — Salt Wind Explorer & Berge Vinson

2023 – 2024

- Salt Wind Explorer: Full project management of the draft increase study — structural assessment, stability analysis, deliverable production, and class and client liaison. Technically complex engagement involving highly detailed class comment responses and full project closure.
- Berge Vinson: Draft increase feasibility study followed by complete delivery package including structural drawings, stability reports, and class-submission documents.

L&T New Drydock Gate & Bombay Drydock Gates — Production Engineering

Cochin Shipyard / Mumbai | 2018 – 2021

- Delivered complete production drawing packages for the L&T New Drydock Gate at Cochin Shipyard — shop drawings, assembly drawings, bill of materials, nesting plans, and full yard support. Contributed to a 15% reduction in potential construction delays.
- Detailed engineering and production drawings for Bombay Drydock Gates.
- Prepared General Arrangement and Structural drawings for floating caisson gates and tugs.

Other Notable Projects

- 3D Structural Modelling: Tugs, LPG Cylinder Carriers, MOPUs, Jack-up Barges, Modular Pontoon Barges, Passenger Ferries, H-Barges, Stern Trawlers (CADMATIC Hull, Rhino 3D, AVEVA PDMS).
- Concept Design: Zero-emission offshore supply vessel; floating bridge across River Tapti; Flettner Rotor design; Fortuna Magpie (3-fuel vessel concept); Decarbonisation feasibility for pilot vessels (Prince Madog).
- CYBERMASTER Programme: Pilot vessel projects and concept design coordination under the Cybermarine CYBERMASTER initiative.
- Design Automation: Developed automation scripts (Python, C#) for production drawing generation, reducing manual effort and improving accuracy in line with Industry 4.0.
- MSc Research (University of Southampton, 2023–24): Numerical investigation of Wind-Assisted Ship Propulsion — Flettner rotor scaling, towing tests, self-propulsion tests, leeway and virtual wind tunnel tests using Siemens Star-CCM+.

TECHNICAL CAPABILITIES

Production Design	Hydrodynamics & Analysis	Simulation & CFD	Project & Modelling
• AutoCAD •CADMATIC Hull & Structure • AVEVA PDMS • Rhino 3D	• Maxsurf • DNV Sesam GeniE •GHS(General Hydrostatics) • Ansys Workbench • HydroD	• Siemens Star-CCM+ • Matlab • Solidworks	• Oracle Primavera • MS Office Suite • Ultimaker Cura • 3D Printing

Coding: Python (Intermediate) | C# (Intermediate)
Classification Society Experience: Bureau Veritas (BV) | DNV | Lloyd's Register (LR) | ABS | IRS
Vessel Types: CSV / SPS | DSV | FPSO / FSO | MOPU / Jack-up | Tug | LPG Carrier | Passenger / Expedition | Bulk Carrier | Aquafarm | Drydock Structures

ACADEMIC QUALIFICATIONS

MSc Maritime Engineering Science (with Merit) *2023 – 2024*
University of Southampton, Southampton, United Kingdom

B.E. Naval Architecture and Offshore Engineering *2014 – 2018*
AMET University (Academy of Maritime Engineering and Science), Chennai, India
Also: Diploma in Business Management — First Class (Dual Degree, 2016–2018)